Midterm review problem set

Instructor solutions

From Euclid to finite fields

16 review problems | 160 practice points + 20 optional | Suggested time: 5–7 hours over several sittings

Preparation. Earlier lectures and assignments, ending with finite-fields Lecture 2: independence, bases, dimension, prime-power cardinalities, and quadratic extensions.

How to use this set. This is a review bank, not a timed mock exam. Try each problem before reading a solution; explain every use of a theorem. Write on separate paper. The point values indicate relative weight. You may quote results proved in class except when explicitly asked to prove them. The final two optional problems revisit the earlier enrichment set. Finite-field cyclicity, matrix counts, Frobenius, and Fibonacci period arguments from Lectures 3–4 are outside this set's scope.

Notation. $v_p(m)$ is the exponent of the prime p in a nonzero integer m; extend it to nonzero rationals by $v_p(a/b) = v_p(a) - v_p(b)$. The functions φ, τ, σ count units, count positive divisors, and sum positive divisors, respectively. The Möbius function satisfies $\mu(1) = 1$, vanishes on integers divisible by a prime square, and equals $(-1)^r$ on a product of r distinct primes.

Instructor note. Each core problem is worth 10 practice points; subpart allocations are shown. Accept equivalent arguments and award method credit after carried arithmetic errors. Suggested time is an estimate, not a classroom-tested completion time. Coverage follows the earlier assignments; the finite-field cutoff follows the main route of Lecture 2.

I. Integers and continued fractions

1. Euclid gives a certificate

[10 points]

(a) Compute $\gcd(252, 165)$ by Euclid and express it as an integer linear combination of 252 and 165. (5)

(b) Find all solutions of $165x \equiv 6 \pmod{252}$, giving the distinct residues modulo 252. Explain why there are exactly that many. (5)

Solution

The successive nonzero remainders are $87, 78, 9, 6, 3$, so the gcd is 3. Back substitution gives $3 = 19 \cdot 252 - 29 \cdot 165$. Multiplying by 2 gives a solution $x = -58$. Dividing the congruence by 3 gives $55x \equiv 2 \pmod{84}$, with a unique solution modulo 84 because $\gcd(55, 84) = 1$. Thus $x \equiv 26 \pmod{84}$, or $x \equiv 26, 110, 194 \pmod{252}$. These three classes exhaust the solutions.

2. What factorization proves

[10 points]

(a) Prove that $\sqrt{72}$ is irrational by comparing prime exponents in a hypothetical equality $a^2 = 72b^2$, with a, b nonzero integers. (4)

(b) Prove there are infinitely many primes congruent to 3 modulo 4. *Hint:* given a proposed complete list $p_1, \ldots, p_r$, consider $N = 4p_1 \cdots p_r - 1$. Explain why some prime divisor of N is 3 modulo 4; do not assume N is prime. (6)

Solution

In $a^2 = 2^3 3^2 b^2$, the exponent of 2 is even on the left and odd on the right, contradicting unique factorization. For the second part, the list is nonempty since 3 is such a prime. The positive odd integer $N \equiv 3 \pmod 4$ has only odd prime factors. If all were $1 \pmod 4$, their product with multiplicities would be $1 \pmod 4$. Thus some prime $q \mid N$ is $3 \pmod 4$. But $N \equiv -1 \pmod{p_i}$ for each i, so q is absent from the list, a contradiction.

3. A finite continued fraction remembers Bézout [10 points]

(a) Find the simple continued fraction of $157/68$ and all of its convergents. Use the convention that the last digit is at least 2. (4)

(b) Multiply the digit matrices $D(a) = \left(\begin{smallmatrix} a & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Use the determinant of their product to find the inverse of 68 modulo 157. (4)

(c) Compute exactly $157/68 - 30/13$. (2)

Solution

Euclid gives $157 = 2\cdot 68 + 21$, $68 = 3\cdot 21 + 5$, $21 = 4\cdot 5 + 1$, $5 = 5\cdot 1$. Thus $157/68 = [2; 3, 4, 5]$, with convergents $2/1, 7/3, 30/13, 157/68$. The product is $\left(\begin{smallmatrix} 157 & 30 \\ 68 & 13 \end{smallmatrix}\right)$, of determinant 1. Consequently $157 \cdot 13 - 68 \cdot 30 = 1$, so $68^{-1} \equiv -30 \equiv 127 \pmod{157}$. Finally $157/68 - 30/13 = (2041 - 2040)/884 = 1/884$.

4. Periodic is not always purely periodic [10 points]

For $x_n = (P_n + \sqrt{D})/Q_n$ and $a_n = \lfloor x_n \rfloor$, recall

$$P_{n+1} = a_n Q_n - P_n, \qquad Q_{n+1} = (D - P_{n+1}^2)/Q_n.$$

(a) Run this recurrence for $\sqrt{11}$ until a state repeats, and give its continued fraction. Justify the floors with integer bounds. (5)

(b) Decide which of $\sqrt{11}$, $3 + \sqrt{11}$, and $(3 + \sqrt{11})/2$ have purely periodic expansions. Use Galois's criterion $x > 1$ and $-1 < x' < 0$, where x' is the conjugate. Read off the purely periodic expansions from (a). (5)

Solution

The states (P_n, Q_n, a_n) are $(0, 1, 3), (3, 2, 3), (3, 1, 6), (3, 2, 3)$. Indeed $3 < \sqrt{11} < 4$, so $3 < (3 + \sqrt{11})/2 < 7/2$ and $6 < 3 + \sqrt{11} < 7$. Thus $\sqrt{11} = [3; \overline{3, 6}]$. Its conjugate is $-\sqrt{11} < -1$, so it is not purely periodic. The other two numbers exceed 1, and their conjugates lie in $(-1, 0)$. Their expansions are $3 + \sqrt{11} = [\overline{6, 3}]$ and $(3 + \sqrt{11})/2 = [\overline{3, 6}]$.

II. Congruences, powers, and arithmetic functions

5. Reassemble residues, and check cancellation [10 points]

(a) Solve $x \equiv 3 \pmod 8$, $x \equiv 4 \pmod 9$, and justify uniqueness modulo 72. (5)

(b) Find every solution of $x^2 \equiv 1 \pmod{15}$ using reduction modulo 3 and 5. Why does the argument "$(x-1)(x+1) = 0$, so one factor is zero" fail modulo 15? (5)

Solution

Set $x = 3 + 8k$. Modulo 9, $3 - k \equiv 4$, so $k \equiv 8$ and $x \equiv 67 \pmod{72}$. A difference of two solutions is divisible by both 8 and 9, hence by 72; adding multiples of 72 preserves both congruences. Modulo each prime $3, 5$, the roots are ± 1. The four CRT combinations yield $1, 4, 11, 14$ modulo 15. Modulo 15 there are nonzero zero divisors: for $x = 4$, both $x - 1 = 3$ and $x + 1 = 5$ are nonzero, but their product is zero.

6. Fermat and inverse pairing [10 points]

(a) Compute $7^{222} \pmod{13}$ efficiently. (3)

(b) Prove Wilson's theorem $(p-1)! \equiv -1 \pmod{p}$ for a prime p by pairing inverses. Account for $p = 2$. (5)

(c) The composite number 341 satisfies $2^{340} \equiv 1 \pmod{341}$. Explain why this is consistent with Fermat's little theorem. (2)

Solution

Since $222 \equiv 6 \pmod{12}$, Fermat gives $7^{222} \equiv 7^6$. Now $7^2 \equiv 10$, $7^3 \equiv 5$, and $7^6 \equiv 25 \equiv 12$. For odd p, each nonzero residue has a unique inverse. The self-inverse ones solve $(a-1)(a+1) = 0$ in $\mathbb{F}_p$, so are exactly $1, -1$. All other factors pair to 1, leaving product -1. For $p = 2$, $1! = 1 \equiv -1$. Fermat gives a necessary condition for primality, not a sufficient one; its converse is false.

7. Orders explain exponent reduction [10 points]

(a) Find the orders of 2 modulo 5, modulo 7, and modulo 35. Justify the relation between the three orders. (4)

(b) Find $\varphi(35)$ and compute $2^{1000} \pmod{35}$. (3)

(c) Explain how Lagrange's theorem applied to $(\mathbb{Z}/n\mathbb{Z})^\times$ proves Euler's theorem. State the hypothesis on the base. (3)

Solution

The powers modulo 5 give order 4; modulo 7, $2, 4, 1$ give order 3. The congruence $2^k \equiv 1 \pmod{35}$ holds precisely when it holds modulo both 5 and 7, or when $4 \mid k$ and $3 \mid k$. Thus the order is 12. Also $\varphi(35) = 4 \cdot 6 = 24$ and $1000 \equiv 4 \pmod{12}$, so $2^{1000} \equiv 16 \pmod{35}$. For $\gcd(a, n) = 1$, the residue of a is in the unit group of size $\varphi(n)$. Its cyclic subgroup has size $\operatorname{ord}_n(a)$, which divides $\varphi(n)$ by Lagrange. Hence $a^{\varphi(n)} \equiv 1 \pmod{n}$.

8. Work one prime at a time [10 points]

(a) Factor 360 and compute $\tau(360), \sigma(360), \varphi(360)$, $\mu(360)$, and $\mu(30)$. Show the prime-power calculations. (5)

(b) Prove that if f is multiplicative, then $F(n) = \sum_{d \mid n} f(d)$ is multiplicative. Explain precisely where coprimality is used. (5)

Solution

Since $360 = 2^3 3^2 5$,

$$\tau(360) = 4 \cdot 3 \cdot 2 = 24, \quad \sigma(360) = 15 \cdot 13 \cdot 6 = 1170, \quad \varphi(360) = 360(1 - 1/2)(1 - 1/3)(1 - 1/5) = 96.$$

Also $\mu(360) = 0$ and $\mu(30) = -1$. For $\gcd(m, n) = 1$, unique factorization gives a bijection between $d \mid mn$ and pairs $a \mid m, b \mid n$, via $d = ab$. Every such pair is coprime, so $f(ab) = f(a)f(b)$. Therefore $F(mn) = \sum_{a \mid m} \sum_{b \mid n} f(a)f(b) = F(m)F(n)$. Also $F(1) = f(1) = 1$, as required by the normalized definition.

9. Invert divisor sums and count exact periods [10 points]

(a) Suppose $\sum_{d\mid n} f(d) = n^2$. Use Möbius inversion to give a formula for $f(n)$, and evaluate $f(12)$. (4)

(b) Let $A(n)$ count labeled binary words of length n with least rotational period exactly n. Explain why $2^n = \sum_{d\mid n} A(d)$. Find $A(6)$ and the number of binary necklaces of length 6 with least period 6, where rotations are identified. Explain the final division. (6)

Solution

Inversion gives $f(n) = \sum_{d\mid n} \mu(d)(n/d)^2$. At 12, only $d = 1, 2, 3, 6$ contribute, giving $144 - 36 - 16 + 4 = 96$. Every length-n word has a unique least period $d \mid n$ and is obtained by repeating a unique length-d word of least period d. Thus the divisor sum holds. Inverting it gives $A(6) = 2^6 - 2^3 - 2^2 + 2 = 54$. A word of least period 6 has six distinct rotations, so there are $54/6 = 9$ such necklaces. Words of smaller period would have smaller rotation classes, which is why the least-period restriction matters.

III. Valuations, primes, and p-adic distance

10. Count prime factors without expanding factorials [10 points]

(a) State Legendre's formula. Use it to find $v_2(40!)$ and the number of trailing decimal zeros of 40!. (3)

(b) Put $B_n = \binom{2n}{n}$. Prove that each summand in

$$v_p(B_n) = \sum_{j\geq 1} \left(\left\lfloor \frac{2n}{p^j} \right\rfloor - 2 \left\lfloor \frac{n}{p^j} \right\rfloor \right)$$

is 0 or 1, and deduce $B_n \mid \operatorname{lcm}(1, \ldots, 2n)$. (4)

(c) For prime p, prove $p \mid \binom{p}{k}$ when $0 < k < p$. Deduce $(X + Y)^p = X^p + Y^p$ as a polynomial identity in characteristic p. (3)

Solution

Legendre gives $v_p(n!) = \sum_{j\geq 1} \lfloor n/p^j \rfloor$. Thus $v_2(40!) = 20 + 10 + 5 + 2 + 1 = 38$, while $v_5(40!) = 8 + 1 = 9$; there are $\min(38, 9) = 9$ trailing zeros. Writing $n/p^j = q + t$, $0 \leq t < 1$, makes each summand $\lfloor 2t \rfloor$, which is 0 or 1. Only powers $p^j \leq 2n$ can contribute, so $v_p(B_n) \leq \max\{j : p^j \leq 2n\}$, the valuation of the lcm (zero if $p > 2n$). Compare valuations for all primes. Finally $p!$ has one factor p and neither $k!$ nor $(p - k)!$ has any. Thus $v_p\binom{p}{k} = 1$. The binomial theorem in characteristic p loses all intermediate coefficients, proving the polynomial identity.

11. Recover the scale of the primes [10 points]

Let $\pi(x) = \#\{p \leq x : p \text{ prime}\}$ and $\vartheta(x) = \sum_{p\leq x} \log p$, with natural logarithms.

(a) Explain why the central binomial coefficient is the largest in its row, and prove $4^n/(2n+1) \leq B_n < 4^n$ for $n \geq 1$. (3)

(b) Use Problem 10 to prove (4)

$$\pi(2n) \geq \frac{2n \log 2 - \log(2n + 1)}{\log(2n)}.$$

(c) You may use the lecture estimate $\vartheta(x) \leq Ax$ for a fixed constant $A > 0$ and all $x \geq 2$. Show $\pi(x) \leq \sqrt{x} + 2Ax/\log x$. Explain why these estimates give $\pi(x) \asymp x/\log x$ but do not give $\pi(x) \sim x/\log x$. (3)

Solution

Consecutive coefficients have ratio $(2n-k)/(k+1)$, so they increase to the center and then decrease. Their sum is 4^n; the largest is at least the average and strictly less than the sum. By Problem 10, B_n divides $L = \operatorname{lcm}(1,\dots,2n)$. For each prime $p \le 2n$, its factor in L is at most $2n$, whence $\log B_n \le \log L \le \pi(2n)\log(2n)$. Combine this with the lower central bound. There are at most $\sqrt{x}$ primes at most $\sqrt{x}$. Each larger prime up to x contributes at least $\frac{1}{2}\log x$ to $\vartheta(x)$, giving the stated upper bound. For general sufficiently large x, set $n = \lfloor x/2 \rfloor$ in the lower bound and use $\pi(x) \ge \pi(2n)$. The resulting lower ratio tends to $\log 2 > 0$; also $\sqrt{x} = o(x/\log x)$. Thus fixed positive lower and upper constants work eventually. Such bounds do not identify the limiting ratio as 1, or even prove that a limiting ratio exists.

12. Close in a different arithmetic [10 points]

For nonzero rational r, define $|r|_3 = 3^{-v_3(r)}$; set $|0|_3 = 0$ and $d_3(x,y) = |x-y|_3$.

(a) Compute $|27/20|_3$ and $d_3(1/4, 7/4)$. (2)

(b) Prove $|x+y|_3 \le \max(|x|_3, |y|_3)$ for rational x, y. Prove equality if $|x|_3 \ne |y|_3$. (4)

(c) Find the 3-adic limit in $\mathbb{Q}$ of $S_n = \sum_{j=0}^{n} 3^j$ by computing the distance to your proposed limit. Does S_n have a finite real limit? (4)

Solution

The values are $1/27$ and $|-3/2|_3 = 1/3$. For nonzero x, y, write $x = 3^a u$, $y = 3^b v$, where u, v are rationals with numerator and denominator prime to 3, and suppose $a \le b$. Then $x + y = 3^a(u + 3^{b-a}v)$ has valuation at least a (or is zero). If $a < b$, putting the parenthesis over a denominator prime to 3 shows its numerator is not divisible by 3, so the valuation is exactly a. Cases involving zero are immediate. Taking 3 to the negative valuation proves both assertions. The geometric sum is $S_n = (3^{n+1}-1)/2$, so the proposed limit is $-1/2$ and $d_3(S_n, -1/2) = |3^{n+1}/2|_3 = 3^{-(n+1)} \to 0$. As real numbers these positive sums grow without bound.

IV. Finite fields: through Lecture 2

13. Units, zero divisors, and characteristic [10 points]

(a) List the units of $\mathbb{Z}/12\mathbb{Z}$ and exhibit two nonzero residues whose product is zero. (2)

(b) Prove that a finite field has prime characteristic. Explain how its multiples of 1 form a copy of $\mathbb{F}_p$. (4)

(c) A finite commutative ring R has $1 \ne 0$ and no nonzero zero divisors. Prove it is a field by considering multiplication by a fixed nonzero element. Where is finiteness used? (4)

Solution

The units are $1, 5, 7, 11$; for instance $2 \cdot 6 = 0$ modulo 12. In a finite field, two distinct nonnegative multiples of 1 coincide, so a positive multiple is zero. Let m be the least positive such integer. If $m = ab$ with $1 < a, b < m$, then $(a \cdot 1)(b \cdot 1) = 0$ has two nonzero factors, impossible in a field. Thus $m = p$ is prime. The map $\mathbb{Z} \to K$, $a \mapsto a \cdot 1$, has kernel $p\mathbb{Z}$ and identifies its image with $\mathbb{F}_p$. For fixed $a \ne 0$ in R, $ax = ay$ implies $a(x-y) = 0$, hence $x = y$. Multiplication by a is injective, and finiteness makes it surjective. In particular, $ab = 1$ for some $b \in R$. This holds for every nonzero a.

14. Why dimension is well defined [10 points]

(a) In $\mathbb{F}_5^2$, show that $u = (1, 2)$ and $v = (2, 1)$ form a basis. Find the coordinates of $(3, 4)$ in this basis. (3)

(b) Let $v_1, \ldots, v_r$ be independent and let $w_1, \ldots, w_m$ span a vector space over a field. Reproduce the exchange argument proving $r \leq m$: after inserting the first k independent vectors, why can the next insertion remove an old w rather than a previously inserted v? Deduce that any two finite bases have the same length. (7)

Solution

If $au + bv = 0$, then $a + 2b = 0$ and $2a + b = 0$, hence $-3b = 0$ in $\mathbb{F}_5$, so $b = a = 0$. The map $(a, b) \mapsto au + bv$ is injective on a set of 25 elements into $\mathbb{F}_5^2$, also of size 25; it is onto, so this is a basis. Also $(3, 4) = 4v$, giving coordinates $(0, 4)$. After k exchanges, the list $v_1, \ldots, v_k$ and $m - k$ old vectors still spans. Write v_{k+1} as a combination of this list. Some old vector has a nonzero coefficient, since otherwise v_{k+1} would lie in the span of $v_1, \ldots, v_k$, contrary to independence. Solve for that old vector by dividing by its coefficient; replacing it by v_{k+1} preserves the span. If $r > m$, after m insertions the spanning list is $v_1, \ldots, v_m$, forcing v_{m+1} into its span, a contradiction. Thus $r \leq m$. Apply the inequality in both directions to two bases to obtain equality of their lengths.

15. Count coordinates, not residue classes [10 points]

(a) Prove every finite field K has p^d elements for some prime p and integer $d \geq 1$. Explain why a basis over $\mathbb{F}_p$ exists and why counting its coordinates counts each element exactly once. (6)

(b) If a field has 81 elements, determine its characteristic and its dimension over its prime field. Can a field have 12 elements? Does the prime-power argument by itself construct a field of every prime-power size? (4)

Solution

By Problem 13, K contains its prime field $\mathbb{F}_p$ and is a vector space over it using field addition and multiplication by scalars in $\mathbb{F}_p$. Start with the empty independent list and repeatedly add a vector outside its span. Such an addition preserves independence: a relation with nonzero coefficient of the new vector would put it in the old span. Each addition strictly enlarges the span, so finiteness forces the procedure to stop at a spanning independent list, a basis. If it has length d, spanning gives existence and independence gives uniqueness of coordinates in $\mathbb{F}_p^d$. Thus $|K| = p^d$; since $1 \neq 0$, $d \geq 1$. For $81 = 3^4$, the characteristic is 3 and dimension 4. The integer 12 is not a prime power, so no field has that cardinality. The argument proves necessity; a general existence theorem has not been proved by this counting argument.

16. Build a field with twenty-five elements [10 points]

Use formal pairs $a + bs$ with $a, b \in \mathbb{F}_5$, and the relation $s^2 = 2$. Addition is coefficientwise and multiplication is distributive with s^2 replaced by 2. You may use the resulting commutative ring laws.

(a) List the squares in $\mathbb{F}_5$. Prove that every nonzero $a + bs$ has inverse $(a - bs)/(a^2 - 2b^2)$, explaining why the denominator is nonzero. (4)

(b) Compute $(1 + s)(2 + 3s)$ and $(1 + s)^{-1}$, and check the inverse by multiplication. (2)

(c) Give a basis over $\mathbb{F}_5$, the cardinality, and the characteristic of this field. Explain why it differs from $\mathbb{Z}/25\mathbb{Z}$. (2)

(d) If we instead imposed $t^2 = 4$ on formal pairs over $\mathbb{F}_5$, would the resulting ring be a field? Give explicit evidence. (2)

Solution

The squares are $0, 1, 4$, so 2 is nonsquare. The product of $a + bs$ with $a - bs$ is $a^2 - 2b^2$. If $b = 0$ and $a \neq 0$, this is nonzero. If $b \neq 0$, its vanishing would give $(a/b)^2 = 2$, a contradiction. Division by this nonzero scalar gives the inverse. The product is $2 + 5s + 3s^2 = 8 = 3$ in $\mathbb{F}_5$. The inverse is $(1 - s)/(1 - 2) = 4 + s$; indeed $(1 + s)(4 + s) = 6 + 5s = 1$. The basis is $1, s$, since formal pairs have unique coefficients. The field has 25 elements and characteristic 5. The ring $\mathbb{Z}/25\mathbb{Z}$ has characteristic 25 and nonzero zero divisor 5. For the alternative construction, $(t - 2)(t + 2) = t^2 - 4 = 0$, but both factors are nonzero formal pairs, so this ring is not a field.

Optional review of earlier enrichment

These two problems review the assignment explicitly labeled optional enrichment; they do not introduce a later finite-fields lecture.

Optional challenge: Inclusion–exclusion as inversion [10 extra points]

In a group of 40 people, the numbers in activities A, B, C are $20, 18, 15$; the pairwise intersection counts are $9, 7, 6$, and the triple intersection count is 3. Pairwise counts include the triple intersection. Find the number in exactly one activity, exactly two activities, and no activities. Explain the alternating signs as recovering exact counts from counts that include additional memberships.

Solution

The counts in only A, B, C are respectively $20 - 9 - 7 + 3 = 7$, $18 - 9 - 6 + 3 = 6$, and $15 - 7 - 6 + 3 = 5$. Thus exactly one has count 18. Exactly two has count $(9 - 3) + (7 - 3) + (6 - 3) = 13$. The union has $20 + 18 + 15 - 9 - 7 - 6 + 3 = 34$, so none has count 6. For instance, subtracting $A \cap B$ and $A \cap C$ from A subtracts the triple intersection twice, so it must be added back once. Suggested credit: exact single counts 4; exact double counts 2; none 2; explanation of inversion 2.

Optional challenge: Interpolation and a finite Fourier transform [10 extra points]

In $\mathbb{F}_5$, let $\omega = 2$, whose powers are $1, 2, 4, 3$. A polynomial f of degree at most 3 satisfies $f(1) = 1$, $f(2) = 0$, $f(4) = 0$, $f(3) = 0$. Recover f using either Lagrange interpolation or

$$c_k = 4^{-1} \sum_{j=0}^{3} f(\omega^j)\omega^{-jk}, \qquad f(X) = \sum_{k=0}^{3} c_k X^k.$$

Check all four values. Why is the degree restriction necessary for uniqueness? Give another polynomial of higher degree with the same values.

Solution

Only $j = 0$ contributes, so each $c_k = 4^{-1} = 4$ in $\mathbb{F}_5$. Thus $f(X) = 4(1 + X + X^2 + X^3)$. At 1 this gives $16 = 1$; at $2, 4, 3$, respectively, the parenthesized sums are $1 + 2 + 4 + 3$, $1 + 4 + 1 + 4$, and $1 + 3 + 4 + 2$, all zero modulo 5. Two degree-at-most-3 polynomials agreeing at four distinct points have difference zero by the polynomial root bound. Without that restriction, $f(X) + (X^4 - 1)$ has the same four values and is different. Suggested credit: coefficients 4; checks 2; uniqueness argument 2; counterexample 2.